Addition of integers

Two rules — same signs and different signs — and a number line that explains both.

LESSON 25–26 | 2 × 40 MIN | MATEMATIKA 7, §10 | S8 1.2

LESSON CLOCK

15' | 25' | 25' | 20' | 5'

On the line	15 min
Same signs	25 min
Different signs	25 min
Several terms	20 min
Homework	5 min

LEARNING OBJECTIVES

- Add two integers with the same sign and with different signs.
- Use the coordinate line as a model for addition.
- Use the commutative and associative laws to add several integers.
- Solve word problems involving gains and losses.

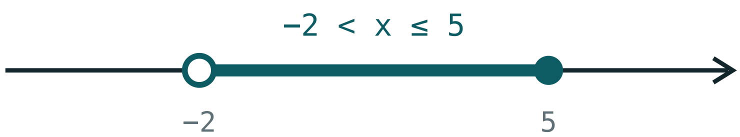
TEXTBOOK

National	pp. 64–70
Cambridge	Stage 8, pp. 6–10
Workbook	Exercise 1.2

01 Explanation

On the line

To add b to a: start at a and move $|b|$ units — right if b is positive, left if it is negative.



open circle = strict · filled = included

Adding a positive number moves right; adding a negative one moves left.

SUM	START	MOVE	END
$3 + 4$	3	4 right	7
$3 + (-4)$	3	4 left	-1
$-3 + 4$	-3	4 right	1
$-3 + (-4)$	-3	4 left	-7

THE LINE IS THE DEFINITION, THE RULES ARE THE SHORTCUT

Anyone who forgets a rule can rebuild it in five seconds on a sketch. That is why the picture comes first in this lesson and the rules second.

Same signs

> Add the moduli and keep the common sign

SUM	MODULI	ANSWER
$5 + 8$	$5 + 8 = 13$	13
$(-5) + (-8)$	$5 + 8 = 13$	-13
$(-12) + (-7)$	$12 + 7 = 19$	-19

Two debts make a bigger debt; two falls in temperature make a bigger fall. The rule matches every situation.

Different signs

> Subtract the smaller modulus from the larger, and keep the sign of the larger modulus

SUM	MODULI	LARGER	ANSWER
$9 + (-4)$	$9, 4$	9	5
$4 + (-9)$	$4, 9$	9	-5
$(-12) + 7$	$12, 7$	12	-5
$6 + (-6)$	$6, 6$	equal	0

THE SIGN COMES FROM THE LARGER MODULUS, NOT FROM THE FIRST NUMBER

$4 + (-9) = -5$, even though the sum starts with a positive. Deciding the sign first, then the size, keeps the two steps separate.

Several terms

Addition is **commutative** and **associative**, so the terms may be reordered and regrouped freely.

$$a + b = b + a \quad (a + b) + c = a + (b + c)$$

Method. Add all the positives, add all the negatives, then combine the two results.

SUM	POSITIVES	NEGATIVES	ANSWER
$7 - 3 + 5 - 12$	12	-15	-3
$-4 + 9 - 6 + 1$	10	-10	0
$-2 - 5 - 8$	0	-15	-15

TWO PILES, ONE SUBTRACTION

Grouping like signs turns a long chain into a single easy step. It is the same idea as collecting like terms in algebra, which arrives later this year.

02 Worked examples

EXAMPLE 1 **Compute $(-5) + (-8)$, $9 + (-4)$ and $4 + (-9)$.**

1. Same signs: add moduli, keep the sign — -13.
2. Different signs: $9 - 4 = 5$, sign of 9 — 5.
3. Different signs: $9 - 4 = 5$, sign of -9 — -5.
4. Sign first, then size.

ANSWER

$-13, 5, -5$

EXAMPLE 2 **Compute $7 - 3 + 5 - 12$.**

1. Positives: $7 + 5 = 12$.
2. Negatives: $-3 - 12 = -15$.
3. $12 + (-15)$
4. $= -3$

ANSWER

-3

EXAMPLE 3

A shop makes a profit of *45 000*, then a loss of *70 000*, then a profit of *30 000*. Find the net result.

1. Profits: $45\,000 + 30\,000 = 75\,000$.
2. Loss: $-70\,000$.
3. $75\,000 + (-70\,000)$
4. $= +5000$ — a small profit.

ANSWER

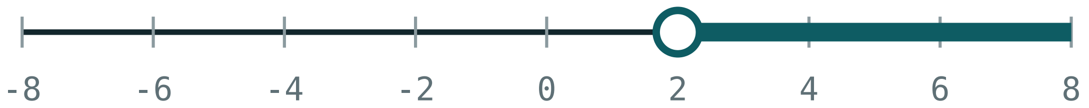
A profit of 5000

03 Interactive model

Do every first example twice — once by the rule and once by walking the line on the floor; the two answers agreeing is what makes the rule trustworthy.

Adding on the line

$$x > 2$$



inequality `x > 2` | boundary `2 (open)` | x = 0 works? **no** | x = 6 works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Addition	Qo‘shish	Сложение
Sum	Yig‘indi	Сумма
Same sign	Bir xil ishorali	Одинаковые знаки
Different signs	Turli ishorali	Разные знаки
Commutative law	O‘rin almashtirish qonuni	Переместительный закон
Associative law	Guruhlash qonuni	Сочетательный закон
Term	Qo‘shiluvchi	Слагаемое
Gain and loss	Foyda va zarar	Прибыль и убыток

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$(-5) + (-8)$ equals:

- 13
- -13
- 3
- -3

$9 + (-4)$ equals:

- 13
- -13
- 5
- -5

$4 + (-9)$ equals:

- 5
- -5
- 13
- -13

$6 + (-6)$ equals:

- 12
- -12
- 0
- 6

For different signs you:

- add the moduli
- subtract the moduli
- multiply them
- ignore them

Addition is:

- commutative only
- associative only
- both
- neither

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $3 + 4$

   ANSWER 7

2. $3 + (-4)$

   ANSWER -1

3. $-3 + 4$

   ANSWER 1

4. $-3 + (-4)$

   ANSWER -7

5. $(-5) + (-8)$

   ANSWER -13

6. $9 + (-4)$

   ANSWER 5

7. $6 + (-6)$

   ANSWER 0

Medium · the standard question

7 PROBLEMS

1. $(-12) + 7$

   ANSWER -5

2. $4 + (-9)$

   ANSWER -5

3. $7 - 3 + 5 - 12$

   ANSWER -3

4. $-4 + 9 - 6 + 1$

   ANSWER 0

5. $-2 - 5 - 8$

   ANSWER -15

6. $15 + (-15) + 8$

   ANSWER 8

7. Profit 45, loss 70, profit 30

   ANSWER $+5$

Hard · stretch and reason

7 PROBLEMS

1. $-17 + 25 - 8 + 3 - 11$

   ANSWER -8

2. $(-3) + (-3) + (-3) + (-3)$

   ANSWER -12

3. A temperature of $-6°$ rises 4, falls 9, rises 12

   ANSWER $+1°$

4. Find x: $x + (-7) = 3$

   ANSWER 10

5. Find x: $-5 + x = -12$

   ANSWER -7

6. The sum of all integers from -10 to 10

   ANSWER 0

7. The sum of all integers from -10 to 12

   ANSWER 23

07 Homework

Homework — 5 tasks

Decide the sign of the answer before computing its size.

1. Compute $(-7) + (-6)$, $11 + (-4)$ and $5 + (-13)$.
2. Compute $9 - 4 + 7 - 15$.
3. Compute $-3 - 8 + 20 - 6$.
4. A bank balance of $60\ 000$ has $85\ 000$ withdrawn and $40\ 000$ paid in. Find the balance.
5. Find x if $x + (-9) = -2$.